

If you connect from your mobile phone to book a call taxi it automatically connects you to the first available cab in your area wherever you are, as long as there's taxi service. What type of coordinate system used to locate and navigate? www.mapbox.com

Example 6.32 If a line joining two points (3, 0) and (5, 2) is rotated about the point (3, 0) in counter clockwise direction through an angle 15°, then find the equation of the line in the new position

Solution:

Let P (3, 0) and Q (5, 2) be the given points.

Slope of $PQ = \dfrac{y_2 - y_1}{x_2 - x_1} = 1 \Rightarrow$ the angle of inclination of the line $PQ = \tan^{-1}(1) = \dfrac{\pi}{4} = 45°$

$\therefore$ The slope of the line in new position is

$$m = \tan(45° + 15°) \Rightarrow \text{ Slope} = \tan(60°) = (\sqrt{3})$$

$\therefore$ Equation of the straight line passing through (3, 0) and with the slope $\sqrt{3}$ is

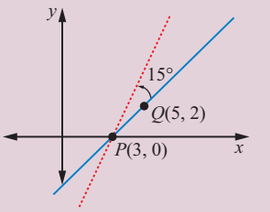


Figure 6.47

$$y - 0 = \sqrt{3}(x - 3)$$
$$\sqrt{3}x - y - 3\sqrt{3} = 0$$

Exercise - 6.3

1. Show that the lines are $3x + 2y + 9 = 0$ and $12x + 8y - 15 = 0$ are parallel lines.
2. Find the equation of the straight line parallel to $5x - 4y + 3 = 0$ and having x-intercept 3.
3. Find the distance between the line $4x + 3y + 4 = 0$, and a point (i) $(-2, 4)$ (ii) $(7, -3)$
4. Write the equation of the lines through the point $(1, -1)$
   (i) parallel to $x + 3y - 4 = 0$
   (ii) perpendicular to $3x + 4y = 6$
5. If $(-4, 7)$ is one vertex of a rhombus and if the equation of one diagonal is $5x - y + 7 = 0$, then find the equation of another diagonal.
6. Find the equation of the lines passing through the point of intersection lines $4x - y + 3 = 0$ and $5x + 2y + 7 = 0$, and (i) through the point $(-1, 2)$ (ii) Parallel to $x - y + 5 = 0$ (iii) Perpendicular to $x - 2y + 1 = 0$
7. Find the equations of two straight lines which are parallel to the line $12x + 5y + 2 = 0$ and at a unit distance from the point (1, − 1).
8. Find the equations of straight lines which are perpendicular to the line $3x + 4y - 6 = 0$ and are at a distance of 4 units from (2, 1).
9. Find the equation of a straight line parallel to $2x + 3y = 10$ and which is such that the sum of its intercepts on the axes is 15.